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ELECTROLYSIS DEVICE

Abstract

An electrolysis cell of an electrolysis device includes a membrane electrode assembly in which an electrolyte membrane is interposed between a first electrode and a second electrode. The membrane electrode assembly is positioned between a first separator and a second separator. The electrolysis device further includes a seal member and a protection member. The protection member surrounds the outer periphery of the second electrode. The protection member includes a first portion and a second portion. The first portion is interposed between the electrolyte membrane and the seal member. The second portion is interposed between the electrolyte membrane and the second separator.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2024-032838 filed on Mar. 5, 2024, the contents of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

Field of the Invention

[0002] The present invention relates to an electrolysis device for performing electrolysis of a fluid.

Description of the Related Art

[0003] As one type of electrolysis device, a water electrolysis device that electrolyzes water to obtain hydrogen and oxygen is known (for example, see JP 2019-157213 A). The water electrolysis device includes an electrolysis cell. The electrolysis cell includes a membrane electrode assembly, and a first separator and a second separator that sandwich the membrane electrode assembly therebetween. The membrane electrode assembly includes a first electrode, a second electrode, and an electrolyte membrane interposed between the first electrode and the second electrode. The first electrode is one of an anode and a cathode, and the second electrode is the other of the anode and the cathode.

[0004] When the electrolyte membrane is a proton conductor, electrons, protons, and oxygen are produced at the cathode, and hydrogen is produced at the anode. The hydrogen is at a higher pressure than the oxygen. When the electrolyte membrane is an anion conductor, hydrogen and hydroxide ions are produced at the cathode, and oxygen, water, and electrons are produced at the anode. The oxygen is at a higher pressure than the hydrogen. In this way, in the water electrolysis device, the high-pressure gas is generated at one of the first electrode and the second electrode.

SUMMARY OF THE INVENTION

[0005] The electrolysis device is provided with a seal member which surrounds the outer periphery of the electrode where the high-pressure gas is generated and which is interposed between the separator and the electrolyte membrane. In some cases, an inner peripheral side wall portion is provided on the inner circumference of the seal member. In this configuration, a surface of the inner peripheral side wall portion that faces the electrolyte membrane may press the electrolyte membrane. Under such a situation, there is a concern that the electrolyte membrane pressed by the inner peripheral side wall portion may be deformed.

[0006] The seal member is deformed by receiving the pressure of the high-pressure gas. Accordingly, the electrolyte membrane is pressed by the deformed seal member. In this case, the electrolyte membrane pressed by the seal member may be deformed.

[0007] An object of the present invention is to solve the aforementioned problems.

[0008] An aspect of the present invention is an electrolysis device including an electrolysis cell including a membrane electrode assembly in which an electrolyte membrane is interposed between a first electrode and a second electrode, and a first separator and a second separator that sandwich the membrane electrode assembly therebetween.

[0009] The electrolysis device includes a fluid supply unit that supplies a fluid involved in an electrolytic reaction, to the first electrode, a power source that applies a voltage between the first electrode and the second electrode, a seal member that surrounds an outer periphery of the second electrode and is interposed between the electrolyte membrane and the second separator, and a protection member that surrounds an outer periphery of the second electrode. The protection member includes a first portion interposed between the electrolyte membrane and the seal member, and a second portion interposed between the electrolyte membrane and the second separator. The first portion and the second portion are different from each other.

[0010] In the protection member, the first portion interposed between the electrolyte membrane and the seal member protects the electrolyte membrane when, for example, the seal member is deformed as a result of receiving the pressure of the high-pressure gas. This prevents the electrolyte membrane from being deformed.
[0011] The above and other objects, features, and advantages of the present invention will become more apparent from the following description when taken in conjunction with the accompanying drawings, in which a preferred embodiment of the present invention is shown by way of illustrative example.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0012] FIG. **1** is a schematic perspective view of an electrolysis device (first water electrolysis device) according to a first embodiment of the present invention;
[0013] FIG. **2** is a cross-sectional view of an electrolysis cell taken along a diameter direction thereof;
[0014] FIG. **3** is a cross-sectional view of a main part of a configuration in which a protection member is provided on an electrolyte membrane such that a rubber sheet faces the electrolyte membrane and a metal sheet faces the seal member;
[0015] FIG. **4** is a cross-sectional view of a main part of a configuration in which a protection member is provided on an electrolyte membrane such that a metal sheet faces the electrolyte membrane and a rubber sheet faces the seal member;
[0016] FIG. **5** is a cross-sectional view of a main part showing a state in which the seal member is deformed;
[0017] FIG. **6** is a sectional view of a main part of an electrolysis device (second water electrolysis device) according to a second embodiment of the present invention; and
[0018] FIG. **7** is a cross-sectional view of a main part showing a state in which the seal member has moved toward an outer peripheral side wall portion.
DETAILED DESCRIPTION OF THE INVENTION
[0019] FIG. **1** is a schematic perspective view of an electrolysis device **200** according to a first embodiment. In the first embodiment, the electrolysis device **200** is a first water electrolysis device **10** that performs electrolysis of water. Therefore, the first water electrolysis device **10** will be described in detail below. However, the electrolysis device **200** is not limited to the first water electrolysis device **10**, as long as it is any device that generates a gas at a second electrode **42***b* shown in FIG. **2**.
[0020] In the first water electrolysis device **10**, as a result of electrolysis of water, a first gas is generated at a first electrode **42***a* shown in FIG. **2**, and a second gas is generated at the second electrode **42***b*. The second gas is set to a higher pressure than the first gas. In the present specification, the second electrode **42***b* refers to an electrode for obtaining a higher-pressure gas. For the sake of simplicity and easy understanding, the first embodiment will exemplify a configuration in which oxygen is generated as the first gas at the first electrode **42***a* and hydrogen is generated as the second gas at the second electrode **42***b*.
[0021] The first water electrolysis device **10** includes an electrolysis cell **12**. As shown in FIG. **1**, in the first water electrolysis device **10**, a plurality of electrolysis cells **12** are stacked in a vertical direction (direction of arrow A), thereby forming a stack body **14**. A terminal plate **16***a*, an insulating plate **18***a*, and an end plate **20***a* are sequentially arranged in the upward direction at one end (upper end) in the stacking direction of the stack body **14**. At the other end (lower end) of the stack body **14** in the stacking direction, a terminal plate **16***b*, an insulating plate **18***b*, and an end plate **20***b* are arranged sequentially in the downward direction. The stacking direction of the

electrolysis cells **12** may be a horizontal direction (direction of arrow B).

[0022] A pipe (not shown) is connected to the end plate **20***a*. The pipe is provided with a back pressure mechanism (not shown) capable of restricting the discharge of hydrogen gas from a hydrogen passage **38***c* described later. The end plate **20***a* and the end plate **20***b* are fastened by tightening via the tie rods **22**. Accordingly, a tightening load acts on the plurality of electrolysis cells **12**.

[0023] Side portions of the terminal plate **16***a* and the terminal plate **16***b* are provided respectively with a terminal portion **24***a* and a terminal portion **24***b* which protrude outward in the diametrical direction. The terminal portions **24***a* and **24***b* are electrically connected to a power source **28** for electrolysis via wirings **26***a* and **26***b*, respectively.

[0024] As shown in FIG. **2**, each of the electrolysis cells **12** includes a substantially disc-shaped membrane electrode assembly **30**, a first separator **32**, and a second separator **34**. The first separator **32** and the second separator **34** sandwich and hold the membrane electrode assembly **30** therebetween. A resin frame member **36** is disposed between the first separator **32** and the second separator **34**. The resin frame member **36** surrounds the outer periphery of the membrane electrode assembly **30**. The gap between the first separator **32** and the resin frame member **36** is sealed by a seal member **37***a*, and the gap between the resin frame member **36** and the second separator **34** is sealed by a seal member **37***b*.

[0025] A fluid supply passage **38***a* is formed at one end of the resin frame member **36** in the radial direction (the direction indicated by the arrow B) and extend in the stacking direction (direction of arrow A). A fluid supply unit **90** is connected to the fluid supply passage **38***a*. The fluid supply unit **90** (see FIG. **1**) supplies the fluid supply passage **38***a* with water as a fluid.

[0026] The other end of the resin frame member **36** in the radial direction (the direction indicated by the arrow B) is provided with a fluid discharge passage **38***b* for discharging unreacted water, and oxygen that is generated based on electrode reaction. As shown in FIG. **1**, a supply joint **92***a* is connected to the resin frame member **36** disposed at the other end (lowermost end) in the stacking direction. The fluid supply port **39***a* of the supply joint **92***a* communicates with the fluid supply passage **38***a* shown in FIG. **2**. As shown in FIG. **1**, a discharge joint **92***b* is connected to the resin frame member **36** disposed at one end (uppermost end) in the stacking direction. The fluid discharge port **39***b* of the discharge joint **92***b* communicates with the fluid discharge passage **38***b* shown in FIG. **2**.

[0027] As shown in FIG. **2**, the electrolysis cell **12** has the hydrogen passage **38***c* that penetrates, along the stacking direction, through a central portion, in the diametrical direction, of the electrolysis cell. Hydrogen generated by electrolysis of water flows through the hydrogen passage **38***c*. The pressure of the hydrogen is increased, for example, to a pressure ranging from 1 MPa to 80 MPa.

[0028] The membrane electrode assembly **30** includes an electrolyte membrane **40**, the first electrode **42***a*, and the second electrode **42***b*. The electrolyte membrane **40**, the first electrode **42***a*, and the second electrode **42***b* are sandwiched between a first current collector **44***a* and a second current collector **44***b*. Each of the electrolyte membrane **40**, the first electrode **42***a*, the second electrode **42***b*, the first current collector **44***a*, and the second current collector **44***b* has a substantially ring shape. In the first embodiment, the first electrode **42***a* is an anode where an oxidation reaction occurs, and the second electrode **42***b* is a cathode where a reduction reaction occurs. The electrolyte membrane **40** is a proton exchange membrane through which protons can move, and is, for example, a hydrocarbon (HC)-based membrane or a fluorine-based membrane.

[0029] A space surrounded by the first separators **32**, the resin frame member **36**, and the electrolyte membrane **40** is a first electrode chamber **45***a*. The first electrode chamber **45***a* accommodates therein a flow path forming member **46** and the first current collector **44***a*. The flow path forming member **46** and the first current collector **44***a* are interposed between the first separator **32** and the electrolyte membrane **40**. The flow path forming member **46** is interposed between the first separator **32** and the first current collector **44***a* in the stacking direction.

[0030] The flow path forming member **46** has an inlet protrusion **46***a* and an outlet protrusion **46***b* on the outer periphery. The inlet protrusion **46***a* and the outlet protrusion **46***b* face each other in the radial direction.

[0031] The inlet protrusion **46***a* is provided with a supply connection path **50***a*. The supply connection path **50***a* communicates with the fluid supply passage **38***a* and a fluid flow path **50***b*. The fluid flow path **50***b* communicates with a plurality of holes **50***c*. The holes **50***c* are open toward the first current collector **44***a*. The outlet protrusion **46***b* is provided with a discharge connection path **50***d*. The discharge connection path **50***d* communicates with the fluid flow path **50***b* and the fluid discharge passage **38***b*.

[0032] A protective sheet member **48** is disposed between the first current collector **44***a* and the first electrode **42***a*. The protective sheet member **48** has a plurality of through holes **48***a* extending along the stacking direction.

[0033] A substantially cylindrical passage body **52** is disposed at the diametrical center of the cell and interposed between the first separator **32** and the electrolyte membrane **40**. The passage body **52** includes an inner cylindrical body **54** and an outer cylindrical body **55** surrounding the outer periphery of the inner cylindrical body **54**. The inner cylindrical body **54** is formed of a porous body in which the hydrogen passage **38***c* is formed. A gap between the inner cylindrical body **54** and the outer cylindrical body **55** is sealed by an O-ring **56***a* and an O-ring **56***b*.

[0034] An annular step portion **55***s* is formed on an end surface of the outer peripheral portion of the outer cylindrical body **55** that faces the electrolyte membrane **40**. An inner peripheral portion of the protective sheet member **48** is inserted into the annular step portion **55***s*.

[0035] A space surrounded by the electrolyte membrane **40**, the resin frame member **36**, and the second separator **34** is a second electrode chamber **45***b*. The second electrode chamber **45***b* accommodates therein the second current collector **44***b* and a load applying mechanism **58**. The second current collector **44***b* and the load applying mechanism **58** are interposed between the electrolyte membrane **40** and the second separator **34**.

[0036] The load applying mechanism **58** includes, for example, a conductive elastic member such as a plate spring **60**. The plate spring **60** applies a load to the second current collector **44***b* via a metal shim member **62**. The load is applied in a direction in which the second current collector **44***b* is pressed toward the second electrode **42***b*, that is, downward in the stacking direction.

[0037] A conductive sheet **66** and an insulating sheet **68** are disposed between the second current collector **44***b* and the shim member **62**. The conductive sheet **66** is formed of, for example, a metal sheet having the hydrogen passage **38***c* disposed at substantially the center in the diametrical direction. The inner and outer diameters of the conductive sheet **66** are substantially equal to the inner and outer diameters of the second current collector **44***b*. A surface of the conductive sheet **66** that faces the second current collector **44***b* has a recess **66***a*. The insulating sheet **68** is accommodated in the recess **66***a*.

[0038] A cylindrical member **70** is disposed radially inward of the load applying mechanism **58**. The cylindrical member **70** is interposed between the conductive sheet **66** and the second separator **34** in the stacking direction. The hydrogen passage **38***c* is formed at the diametrical center of the cylindrical member **70**. A hydrogen discharge channel **71** is formed in one end surface of the cylindrical member **70** that faces the second separator **34**. The hydrogen discharge channel **71** establishes communication between the second electrode chamber **45***b* and the hydrogen passage **38***c*.

[0039] In the stacking direction, a seal member **80** and a protection member **82** are interposed between the electrolyte membrane **40** and the second separator **34**. The seal member **80** is sandwiched between the protection member **82** and the second separator **34**. The protection member **82** includes a first portion **82***a* interposed between the electrolyte membrane **40** and the seal member **80**, and a second portion **82***b* interposed between the electrolyte membrane **40** and the second separator **34** via an outer peripheral side wall portion **84**.

[0040] The first water electrolysis device **10** has the outer peripheral side wall portion **84** and an inner peripheral side wall portion **86**. The outer peripheral side wall portion **84** surrounds the seal member **80**. Therefore, the inner periphery of the outer peripheral side wall portion **84** faces the outer periphery of the seal member **80**. The outer peripheral side wall portion **84** is formed of, for example, a pressure-resistant member that is separate from the second separator **34** and has a ring shape. The seal member **80** surrounds the inner peripheral side wall portion **86**. Therefore, the inner periphery of the seal member **80** faces the outer periphery of the inner peripheral side wall portion **86**. The inner peripheral side wall portion **86** is, for example, an annular projection projecting downward from the lower surface of the second separator **34**. In this case, the inner peripheral side wall portion **86** is a portion of the second separator **34**.

[0041] The outer peripheral side wall portion **84** may be an annular projection projecting downward from the lower surface of the second separator **34**. The inner peripheral side wall portion **86** may be formed of a ring-shaped member that is separate from the second separator **34**.

[0042] The outer peripheral side wall portion **84** and the inner peripheral side wall portion **86** may be provided on the same member. In this case, the member has, for example, a ring-shaped portion that is separate from the second separator **34**. The outer peripheral side wall portion **84** is provided so as to protrude downward from the lower surface of the outer peripheral end of the ring-shaped portion. The inner peripheral side wall portion **86** is provided to protrude downward from the lower surface of the inner peripheral end of the ring-shaped portion.

[0043] In any of the configurations, an annular groove **88** is formed between the outer peripheral side wall portion **84** and the inner peripheral side wall portion **86**. The seal member **80** is inserted into the annular groove **88**. As shown in FIG. **2**, in the protection member **82**, an inner peripheral end **82***i* of the first portion **82***a* is located inward of an inner peripheral end **86***i* of the inner peripheral side wall portion **86**. The second electrode chamber **45***b* surrounded by the inner peripheral side wall portion **86** and the second separator **34** is formed inward of the inner peripheral side wall portion **86**. The second electrode chamber **45***b* accommodate therein the second electrode **42***b*, the second current collector **44***b*, the conductive sheet **66**, the insulating sheet **68**, the shim member **62**, and the load applying mechanism **58**.

[0044] The protection member **82** is, for example, a single annular sheet. In the protection member **82** having an annular shape, the inner peripheral surface of the first portion **82***a* is close to the outer peripheral portion **42***bo* of the second electrode **42***b*. The inner peripheral surface of the first portion **82***a* may abut on the outer peripheral portion **42***bo* of the second electrode **42***b*. The protection member **82** may has a configuration of not being interposed between the electrolyte membrane **40** and the second electrode **42***b*.

[0045] Suitable examples of the sheet include a metal sheet or a rubber sheet. However, the protection member **82** is not limited to a metal sheet or a rubber sheet. The protection member **82** may be formed of carbon paper.

[0046] When the first water electrolysis device **10** is manufactured, the end plates **20***a* and **20***b* are fastened together by tightening via the tie rods **22** as described above. Accordingly, a tightening load acts on the plurality of electrolysis cells **12**. When the protection member **82** is formed of a metal sheet, the tightening load is moderated. Therefore, the tightening load can be made uniform in the plurality of electrolysis cells **12**.

[0047] When the protection member **82** is formed of a rubber sheet, the tightening load can be made uniform in the plurality of electrolysis cells **12** in the same manner as described above. The rubber sheet is easily compressed when the first water electrolysis device **10** is tightened. Therefore, in each of the plurality of electrolysis cells **12**, the pressure distribution in the plane direction can be equalized. As a result, the cell voltage in each of the plurality of electrolysis cells **12** is stabilized at a low value. Furthermore, the difference in voltage between the plurality of electrolysis cells **12** is reduced.

[0048] As shown in FIGS. **3** and **4**, the protection member **82** may be a laminated sheet **85**. A suitable example of the laminated sheet **85** is a laminate body of a metal sheet **83***a* and a rubber sheet **83***b*. In this case, in the configuration shown in FIG. **3**, the rubber sheet **83***b* faces the electrolyte membrane **40**, and the metal sheet **83***a* faces the seal member **80**. In the configuration shown in FIG. **4**, the metal sheet **83***a* faces the electrolyte membrane **40**, and the rubber sheet **83***b* faces the seal member **80**.
[0049] In the configuration in which the rubber sheet **83***b* faces the electrolyte membrane **40** and the metal sheet **83***a* faces the seal member **80** (see FIG. **3**), even if the metal sheet **83***a* has minute irregularities, the rubber sheet **83***b* is deformed to fill the irregularities, so that the protection member **82** is in close contact with the electrolyte membrane **40**. Therefore, the bonding strength of the protection member **82** to the electrolyte membrane **40** is high.
[0050] In the configuration in which the metal sheet **83***a* faces the electrolyte membrane **40** and the rubber sheet **83***b* faces the seal member **80** (see FIG. **4**), the protection member **82** is also in close contact with the electrolyte membrane **40** in the same manner as described above. In addition, the electrolyte membrane **40** and the second separator **34** are effectively insulated from each other. Further, since the rubber sheet **83***b* is in abutment with the second separator **34**, the second separator **34** is protected from corrosion when the second separator **34** is made of metal.
[0051] In the first embodiment, the inner peripheral end **82***i* of the protection member **82** (first portion **82***a*) is located inward of the inner peripheral end **86***i* of the inner peripheral side wall portion **86**. Thus, as a result of the above-described tightening, a pressing force is applied to the protection member **82** from the lower surface of the inner peripheral side wall portion **86** facing the electrolyte membrane **40**. At this time, the protection member **82** absorbs the pressing force. Therefore, the pressing force acting on the electrolyte membrane **40** is reduced, and thus the deformation of the electrolyte membrane **40** is suppressed.
[0052] Next, the operation of the first water electrolysis device **10** will be described.
[0053] A voltage is applied from the power source **28** to the terminal portion **24***a* of the terminal plate **16***a* and the terminal portion **24***b* of the terminal plate **16***b* shown in FIG. **1**. Water as a fluid is supplied from the fluid supply unit **90**. The water flows into the fluid supply passage **38***a* (see FIG. **2**) of the electrolysis cell **12** through the fluid supply port **39***a*. The water flows through the fluid supply passage **38***a* and the supply connection path **50***a* in the electrolysis cell **12**, and then flows into the fluid flow path **50***b* of the flow path forming member **46**. Then, the water is supplied from the plurality of holes **50***c* to the first current collector **44***a*.
[0054] The water is electrolyzed at the first electrode **42***a*. As a result, protons, electrons, and oxygen are generated. That is, the water is involved in the electrode reaction (oxidation reaction) at the first electrode **42***a*. The protons move to the second electrode **42***b* through the electrolyte membrane **40**, and are combined with the electrons in the second electrode **42***b*. As a result, hydrogen is produced. The hydrogen is discharged from the second electrode chamber **45***b* to the hydrogen passage **38***c* through the pores of the second current collector **44***b* and the hydrogen discharge channel **71**.
[0055] The back pressure mechanism restricts the discharge of the hydrogen from the hydrogen passage **38***c*. Therefore, when the electrolysis reaction of water proceeds in the electrolysis cell **12**, the internal pressure of the second electrode chamber **45***b* increases due to the produced hydrogen. As a result, the internal pressure of the second electrode chamber **45***b* becomes higher than the internal pressure of the first electrode chamber **45***a*, and the hydrogen in the hydrogen passage **38***c* is maintained at a high pressure. Thus, the high-pressure hydrogen whose pressure has been increased to a predetermined pressure can be taken out from the first water electrolysis device **10**. On the other hand, oxygen produced by the electrode reaction (reduction reaction) at the first electrode **42***a* is entrained in the unreacted water and discharged to the outside of the first water electrolysis device **10** through the fluid discharge passage **38***b* and the fluid discharge port **39***b* at normal pressure.

[0056] The high-pressure hydrogen in the second electrode chamber **45***b* flows into the annular groove **88**. Therefore, as shown in FIG. **5**, the seal member **80** may be pressed by the high-pressure hydrogen and may be pressed against the inner peripheral surface of the outer peripheral side wall portion **84**. Further, the seal member **80** is pressed by the high-pressure hydrogen and is deformed so as to expand along the stacking direction.
[0057] As the seal member **80** is deformed as described above, the lower portion of the seal member **80** facing the protection member **82** and the electrolyte membrane **40** presses the protection member **82** such that the protection member **82** extends in the direction orthogonal to the stacking direction. Therefore, the pressing force from the seal member **80** is relaxed by the protection member **82**. Thus, the protection member **82** extends from the second electrode chamber **45***b* where the high-pressure hydrogen is generated to the annular groove **88** into which the high-pressure hydrogen flows. As a result, deformation of the electrolyte membrane **40** is suppressed.
[0058] Since the deformation of the electrolyte membrane **40** is suppressed, the increase in the amount of permeation to the first electrode **42***a* of the high-pressure hydrogen generated in the second electrode **42***b* is suppressed. Therefore, the amount of hydrogen that is collected through the hydrogen passage **38***c* is prevented from being reduced. In addition, hydrogen does not interfere with the electrode reaction at the first electrode **42***a*, thus a decrease in reaction efficiency is avoided. For the reasons described above, sufficient amounts of hydrogen and oxygen can be obtained by the electrolysis of water.
[0059] The effects of the first embodiment are summarized as follows.
[0060] As shown in FIG. **2**, the first water electrolysis device **10** according to the first embodiment includes the protection member **82** interposed between the electrolyte membrane **40** and the seal member **80**. The first portion **82***a* of the protection member **82** suppresses deformation of the electrolyte membrane **40** when the seal member **80** is deformed due to generation of hydrogen at the second electrode **42***b*.
[0061] As a result, the progress of the electrode reaction in the first electrode **42***a* or the second electrode **42***b* is prevented from being reduced due to the deformation of the electrolyte membrane **40**. Further, it is possible to suppress a decrease in the collected amount of the generated high-pressure hydrogen. Thus, sufficient amounts of hydrogen and oxygen can be obtained by the electrolysis of water.
[0062] The inner peripheral surface of the protection member **82** abuts on the outer peripheral portion **42***bo* of the second electrode **42***b*.
[0063] The end surface of the electrolyte membrane **40** facing the seal member **80** is mostly covered with the protection member **82**. Therefore, the deformation of the electrolyte membrane **40** is further suppressed.
[0064] In a configuration, the protection member **82** is made of a metal sheet.
[0065] In this case, the tightening load applied when the first water electrolysis device **10** is tightened is relaxed by the protection member **82**. Therefore, the tightening load can be made uniform in the plurality of electrolysis cells **12**.
[0066] In another configuration, the protection member **82** is made of a rubber sheet.
[0067] In this configuration also, the tightening load can be made uniform in the plurality of electrolysis cells **12** as in the above case. In addition, the protection member **82** made of a rubber sheet is easily compressed when the first water electrolysis device **10** is tightened. Therefore, in each of the plurality of electrolysis cells **12**, the pressure distribution in the plane direction can be equalized. As a result, in each of the plurality of electrolysis cells **12**, the cell voltage is stabilized at a low value. Furthermore, the difference in voltage between the plurality of electrolysis cells **12** is reduced.
[0068] In another configuration, the protection member **82** has a laminated sheet **85** of a metal sheet **83***a* and a rubber sheet **83***b*.
[0069] In this configuration, as shown in FIG. **3**, when the rubber sheet **83***b* faces the electrolyte

membrane **40** and the metal sheet **83***a* faces the seal member **80**, the electrolyte membrane **40** and the second separator **34** are insulated from each other. Further, since the rubber sheet **83***b* is interposed between the metal sheet **83***a* and the electrolyte membrane **40**, metal of the metal sheet **83***a* is prevented from mixing into the electrolyte membrane **40**.

[0070] On the contrary, as shown in FIG. **4**, when the metal sheet **83***a* faces the electrolyte membrane **40** and the rubber sheet **83***b* faces the seal member **80**, the electrolyte membrane **40** and the second separator **34** are insulated from each other in the same manner as described above. Further, since the rubber sheet **83***b* is in abutment with the second separator **34**, the second separator **34** is protected from corrosion when the second separator **34** is made of metal.

[0071] The above effects are also obtained in a second embodiment described later. In addition, the first embodiment has the following unique effects.

[0072] The first water electrolysis device **10** includes an outer peripheral side wall portion **84** facing the outer periphery of the seal member **80**, and an inner peripheral side wall portion **86** facing the inner periphery of the seal member **80** and having the outer periphery surrounded by the seal member **80**. The annular groove **88** is formed between the outer peripheral side wall portion **84** and the inner peripheral side wall portion **86**. The seal member **80** is inserted into the annular groove **88**. The inner peripheral end **82***i* of the protection member **82** is located inward of the inner peripheral end **86***i* of the inner peripheral side wall portion **86**.

[0073] When the stack body **14** is tightened, the pressing force acting in the direction from the lower surface of the inner peripheral side wall portion **86** toward the electrolyte membrane **40** is absorbed by the protection member **82**. Therefore, in this case as well, deformation of the electrolyte membrane **40** is suppressed.

[0074] Next, a second water electrolysis device **100** according to a second embodiment will be described with reference to FIG. **6**. The second water electrolysis device **100** is a device that performs electrolysis of water, similarly to the first water electrolysis device **10**. Moreover, it should be noted that the same constituent elements as those shown in FIGS. **1** to **5** are designated by the same reference numerals and detailed description of such elements will be omitted.

[0075] The second water electrolysis device **100** does not include the inner peripheral side wall portion **86**. Therefore, the seal member **80** surrounds the outer peripheral surface of the second electrode **42***b*. A space **102** communicating with the second electrode chamber **45***b* is formed between the outer peripheral surface of the second electrode **42***b* and the inner peripheral surface of the outer peripheral side wall portion **84**. In the space **102**, the seal member **80** is movable along the diameter direction.

[0076] In the second embodiment, the inner peripheral end **82***i* of the protection member **82** is located inward of an inner peripheral end **80***i* of the seal member **80**. That is, the inner peripheral end **82***i* of the protection member **82** extends into the space **102**. Therefore, the upper surface of the protection member **82** is one surface that forms the space **102**. The inner peripheral surface of the annular protection member **82** abuts on the outer peripheral portion of the second electrode **42***b*.

[0077] In the second embodiment, when high-pressure hydrogen is generated in the second electrode **42***b*, the high-pressure hydrogen is temporarily stored in the space **102**. The high-pressure hydrogen in the space **102** pushes the seal member **80** toward the outer peripheral side wall portion **84**. As a result, as shown in FIG. **7**, the seal member **80** is moved toward the inner peripheral surface of the outer peripheral side wall portion **84** and compressed. At this time also, as in the first embodiment, the pressing force from the lower portion of the seal member **80** is absorbed (relaxed) by the protection member **82**.

[0078] In addition, the exposed area of the upper surface of the protection member **82** increases in accordance with the above-described deformation of the seal member **80**. Therefore, the pressure of the high-pressure hydrogen mainly acts on the upper surface of the protection member **82**. Therefore, the protection member **82** is prevented from peeling off from the electrolyte membrane **40**. Therefore, the electrolyte membrane **40** is effectively protected by the protection member **82**.

Even in the above-described configuration where the inner peripheral side wall portion **86** is not provided, the protection member **82** can prevent the electrolyte membrane **40** from being deformed. Therefore, in the second embodiment also, hydrogen and oxygen can be obtained in sufficient amounts by electrolysis of water.

[0079] The second embodiment exhibits the following advantageous effects.

[0080] The second water electrolysis device **100** includes the outer peripheral side wall portion **84** facing the outer periphery of the seal member **80**. The space **102** is formed between the outer peripheral surface of the second electrode **42***b* and the inner peripheral surface of the outer peripheral side wall portion **84**. The seal member **80** is movably accommodated in the space **102**. The inner peripheral end **82***i* of the protection member **82** is located inward of the inner peripheral end **80***i* of the seal member **80**.

[0081] When the seal member **80** is moved toward the outer peripheral side wall portion **84** by the high-pressure hydrogen stored in the space **102**, the exposed area of the upper surface of the protection member **82** increases. Therefore, the pressure of the high-pressure hydrogen mainly acts on the upper surface of the protection member **82**. Therefore, the protection member **82** is prevented from peeling off from the electrolyte membrane **40**.

[0082] In the first embodiment and the second embodiment, oxygen is generated as the first gas at the first electrode **42***a*, and hydrogen is generated as the second gas at the second electrode **42***b*. However, there can be another configuration in which hydrogen is generated as the first gas at the first electrode **42***a*, and oxygen is generated as the second gas at the second electrode **42***b*. The latter configuration will be briefly described.

[0083] When the electrolyte membrane **40** is made of an anion conductor, a reduction reaction for producing hydrogen and hydroxide ions from water occurs at the first electrode **42***a*. The hydroxide ions are conducted through the electrolyte membrane **40** and move to the second electrode **42***b*. In the second electrode **42***b*, an oxidation reaction occurs in which oxygen, water, and electrons are produced from hydroxide ions. The pressure of the oxygen is increased to a predetermined pressure by the back pressure mechanism. In this way, the first and second water electrolysis devices **10** and **100** can produce hydrogen as the first gas at the first electrode **42***a* and can produce oxygen at a high pressure as the second gas at the second electrode **42***b*.

[0084] As described above, the electrolysis device **200** according to the present invention is not limited to the first water electrolysis device **10** and the second water electrolysis device **100** that perform electrolysis of water. That is, the present invention can be applied to the electrolysis device **200** that performs electrolysis of a substance (fluid) other than water.

[0085] The following Supplementary Notes are further disclosed in relation to the above embodiments.

Supplementary Note 1

[0086] The electrolysis device (**200**) of the present disclosure includes the electrolysis cell (**12**) including the membrane electrode assembly (**30**) in which the electrolyte membrane (**40**) is interposed between the first electrode (**42***a*) and the second electrode (**42***b*), and the first separator (**32**) and the second separator (**34**) that sandwich the membrane electrode assembly therebetween. The electrolysis device further includes the fluid supply unit (**90**) that supplies a fluid involved in the electrolytic reaction, to the first electrode, the power source (**28**) that applies a voltage between the first electrode and the second electrode, the seal member (**80**) that surrounds the outer periphery of the second electrode and is interposed between the electrolyte membrane and the second separator, and the protection member (**82**) that surrounds the outer periphery of the second electrode. The protection member surrounds the outer periphery of the second electrode, and includes the first portion (**82***a*) interposed between the electrolyte membrane and the seal member, and the second portion (**82***b*) interposed between the electrolyte membrane and the second separator. The first portion and the second portion are different from each other.

[0087] When the seal member is deformed as a result of receiving the pressure of the high-pressure gas, the protection member **82** absorbs (relaxes) the pressing force acting in the direction from the deformed seal member toward the electrolyte membrane. This suppresses deformation of the electrolyte membrane. Therefore, the progress of the electrode reaction in the second electrode is prevented from being reduced due to the deformation of the electrolyte membrane. Therefore, a sufficient amount of gas can be obtained at the second electrode by the electrolysis.

Supplementary Note 2

[0088] The electrolysis device according to Supplementary Note 1 may further include the outer peripheral side wall portion (**84**) facing the outer periphery of the seal member; the inner peripheral side wall portion (**86**) facing the inner periphery of the seal member and including the outer periphery surrounded by the seal member; and the annular groove (**88**) formed between the outer peripheral side wall portion and the inner peripheral side wall portion, and the seal member may be inserted into the annular groove, and the inner peripheral end (**82***i*) of the protection member may be located inward of the inner peripheral end (**86***i*) of the inner peripheral side wall portion.

[0089] When the electrolysis device is tightened, the pressing force acting in the direction from the inner peripheral side wall portion toward the electrolyte membrane is absorbed (relaxed) by the protection member. That is, in this case as well, deformation of the electrolyte membrane is avoided by the protection member.

Supplementary Note 3

[0090] The electrolysis device according to Supplementary Note 1 may further include the outer peripheral side wall portion (**84**) facing the outer periphery of the seal member; and the space (**102**) formed between the outer peripheral surface of the second electrode and the inner peripheral surface of the outer peripheral side wall portion and configured to movably accommodate the seal member, and the inner peripheral end (**82***i*) of the protection member may be located inward of the inner peripheral end (**80***i*) of the seal member.

[0091] When the high-pressure gas is generated at the second electrode, the high-pressure gas in the space pushes the seal member toward the outer peripheral side wall portion. Accordingly, the seal member moves toward the inner peripheral surface of the outer peripheral side wall portion, and thus the exposed area of the upper surface (surface facing the space) of the protection member increases.

[0092] Therefore, the pressure of the high-pressure gas mainly acts on the upper surface of the protection member. Therefore, the protection member is prevented from being peeled off from the electrolyte membrane. Therefore, the electrolyte membrane is protected by the protection member.

Supplementary Note 4

[0093] In the electrolysis device according to any one of Supplementary notes 1 to 3, the inner peripheral surface of the protection member may abut on the outer peripheral portion (**42***bo*) of the second electrode.

[0094] In the electrolyte membrane, most of the end surface facing the seal member is covered with the protection member. Therefore, the deformation of the electrolyte membrane is further suppressed.

Supplementary Note 5

[0095] In the electrolysis device according to any one of Supplementary notes 1 to 4, the protection member may be made of a metal sheet.

[0096] According to this configuration, the tightening load applied to the plurality of electrolysis cells can be made uniform.

Supplementary Note 6

[0097] In the electrolysis device according to any one of Supplementary notes 1 to 4, the protection member may be made of a rubber sheet.

[0098] In this configuration also, the tightening load can be made uniform in the plurality of electrolysis cells as in the above. In addition, the protection member made of a rubber sheet is easily compressed when the electrolysis device is tightened. Therefore, in each of the plurality of electrolysis cells, the pressure distribution in the plane direction can be equalized. As a result, in each of the plurality of electrolysis cells, the cell voltage is stabilized at a low value. Furthermore, the difference in voltage between the plurality of electrolysis cells is reduced.

Supplementary Note 7

[0099] In the electrolysis device according to any one of Supplementary Notes 1 to 4, the protection member may include the laminated sheet (**85**) of the metal sheet (**83***a*) and the rubber sheet (**83***b*).

[0100] In the configuration in which the rubber sheet faces the electrolyte membrane and the metal sheet faces the seal member, the electrolyte membrane and the second separator can be insulated from each other. Further, metal is prevented from being mixed into the electrolyte membrane from the metal sheet.

[0101] Similarly, also in the configuration in which the metal sheet faces the electrolyte membrane and the rubber sheet faces the seal member, the electrolyte membrane and the second separator can be insulated from each other. Further, when the second separator is made of metal, the second separator can be protected from corrosion by the rubber sheet.

Supplementary Note 8

[0102] The electrolysis device according to any one of Supplementary notes 1 to 7 may be the water electrolysis device (**10**, **100**) configured to perform electrolysis of water supplied as the fluid, and a gas generated at the second electrode may have a higher pressure than a gas generated at the first electrode.

[0103] In this case, hydrogen and oxygen can be obtained. As described above, when the electrolyte membrane is a proton conductor (H.sup.+ conductor), the gas generated at the first electrode is oxygen, and the gas generated at the second electrode (high-pressure gas) is hydrogen. In contrast, when the electrolyte membrane is an anion conductor (OH.sup.− conductor), the gas generated at the first electrode is hydrogen, and the gas generated at the second electrode (high-pressure gas) is oxygen.

[0104] Moreover, it should be noted that the present invention is not limited to the disclosure described above, but various configurations may be adopted therein without departing from the essence and gist of the present invention.

Claims

1. An electrolysis device comprising: an electrolysis cell including a membrane electrode assembly in which an electrolyte membrane is interposed between a first electrode and a second electrode, and a first separator and a second separator that sandwich the membrane electrode assembly therebetween; a fluid supply unit configured to supply a fluid involved in an electrolytic reaction, to the first electrode; a power source configured to apply a voltage between the first electrode and the second electrode; a seal member surrounding an outer periphery of the second electrode and interposed between the electrolyte membrane and the second separator; and a protection member that surrounds an outer periphery of the second electrode, the protection member including a first portion interposed between the electrolyte membrane and the seal member, and a second portion different from the first portion and interposed between the electrolyte membrane and the second separator.

2. The electrolysis device according to claim 1, further comprising: an outer peripheral side wall portion facing an outer periphery of the seal member; an inner peripheral side wall portion facing an inner periphery of the seal member and including an outer periphery surrounded by the seal member; and an annular groove formed between the outer peripheral side wall portion and the inner peripheral side wall portion, wherein the seal member is inserted into the annular groove, and an inner peripheral end of the protection member is located inward of an inner peripheral end of the inner peripheral side wall portion.

3. The electrolysis device according to claim 1, further comprising: an outer peripheral side wall portion facing an outer periphery of the seal member; and a space formed between an outer peripheral surface of the second electrode and an inner peripheral surface of the outer peripheral side wall portion and configured to movably accommodate the seal member, wherein an inner peripheral end of the protection member is located inward of an inner peripheral end of the seal member.

4. The electrolysis device according to claim 1, wherein an inner peripheral surface of the protection member abuts on an outer peripheral portion of the second electrode.

5. The electrolysis device according to claim 1, wherein the protection member is made of a metal sheet.

6. The electrolysis device according to claim 1, wherein the protection member is made of a rubber sheet.

7. The electrolysis device according to claim 1, wherein the protection member includes a laminated sheet of a metal sheet and a rubber sheet.

8. The electrolysis device according to claim 7, wherein the metal sheet and the rubber sheet are laminated in a manner so that the rubber sheet faces the electrolyte membrane and the metal sheet faces the seal member.

9. The electrolysis device according to claim 7, wherein the metal sheet and the rubber sheet are laminated in a manner so that the metal sheet faces the electrolyte membrane and the rubber sheet faces the seal member.

10. The electrolysis device according to claim 1, wherein the electrolysis device is a water electrolysis device configured to perform electrolysis of water supplied as the fluid, and a gas generated at the second electrode has a higher pressure than a gas generated at the first electrode.